

Public element sets against reconstructed orbits: a record-controlled reference benchmark

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This version, v2, supersedes v1 at doi [10.5281/zenodo.22656992](https://doi.org/10.5281/zenodo.22656992). It corrects manoeuvre exclusions, horizon criteria and covariance coordinates, and adds the completed calendar replay and topocentric comparison while retaining the learned-propagator negative findings.

Evidence object: [benchmark-v2.json](#), SHA-256 `f19c24451e65d8e0baf6b803f379380aab72146e5ef73687b818397bcb5609e5`.

Abstract

Public element-set errors vary sharply with altitude, disturbance and mission. For spacecraft at 460-507 km, last-pass/first-fail brackets are 120/144 hours in April 2024 control, 48/72 in May 2024, 48/72 in August 2024 held out and 24/36 in October 2024 held out. Whole-set deletion leaves these brackets unchanged; spacecraft deletion preserves all except the August 2024 held out bracket. Bands above 850 km pass through the longest tested lead, with mission-dependent fragility. Consistency covariance fails to cover storm residuals and Sentinel-6A even in the control. The September replay gives mean absolute prediction error 4.284 km against 17.254 km for zero prediction; 11 of 13 events improve, with SWOT and one Sentinel-3A event as the exceptions. Constructed topocentric crossings agree in 98.0% of cases away from the deliberate boundary offsets and 52.1% at them. Both learned-propagator checkpoints are rejected. These results describe inspected populations, not transferable operating guarantees.

Population and comparison method

Building on [Flohner, Krag and Klinkrad \(2008\)](#), who estimated covariance from element-set consistency, this work measures component coverage against reconstructed orbits with covariance and residuals expressed in the same basis. Extending [Mason's \(2013\)](#) NASA Ames comparison of element sets with reference ephemerides, it reports mission-specific storm and control comparisons, recorded-manoevre exclusions and deletion-sensitive horizon brackets. Alongside [Parker and Linares \(2024\)](#), who examined catalogue decay and geomagnetic forecast performance during the Gannon storm, it measures propagation residuals against reconstructed orbits without interpreting observed storm conditions as issued forecasts. Following [Acciarini, Baydin and Izzo \(2025\)](#), who introduced differentiable and learned SGP4 propagation, it records paired negative findings for the local checkpoints and public-data recipe rather than claiming to reproduce their experiment.

We compare public SGP4 element sets with reconstructed reference orbits for 15 spacecraft in 5 altitude bands. The sample spans 460-1,338 km and contains 1,249 raw element sets. Of these, 1,193 contribute a usable comparison at some lead, yielding 10,310 set/lead pairs. Each pair compares propagation from an element-set epoch with the reference state at the same time. These are epoch-based reconstructions; state epochs do not establish historical publication or availability.

The windows are April 2024 control, May 2024, August 2024 held out and October 2024 held out. All have been inspected. The retained “held out” names describe their historical roles; the corrected comparisons are descriptive, and none of these windows remains an untouched evaluation population. The supplement records the exact epoch selections and mission composition at each lead.

Published manoeuvre records govern exclusions where integrated. A recorded burn from 24 hours before the element-set epoch through the comparison time excludes that pair. This span is an analysis rule, not a measured catalogue fitting arc. Missing record coverage is not evidence of no burn. Orbit and element-set detectors provide cross-checks on recorded missions and explicitly identified exclusions elsewhere. The IDS registry does not independently establish exhaustive reporting of every firing.

The reference products come from the ESA Swarm, JPL/GFZ GRACE-FO, CNES/SSALTO and Sentinel precise-orbit archives cited in the supplement. GRACE-FO adds another mission, producer and thruster record, although these and Swarm remain related GNSS-derived references. Laser ranging supplies a different measurement on sampled passes. Agreement in range helps check conventions and reference points; it does not measure the complete spatial tail distribution. Debris, eccentric orbits and unmeasured catalogue objects are outside this population.

Residual horizons

A lead passes when at least 95% of finite, usable absolute in-track residuals lie within 25 km. The bracket joins the last passing bin to the first failing bin; it does not locate a continuous transition between them. Passing through the longest tested lead, 168 hours, is right censoring. Missing observations end the supported interval separately from a measured threshold failure. Empirical counts determine the primary result; linear and inverted-CDF quantiles remain alongside them in the supplement.

Band	Window	Primary observed endpoints	Remove one spacecraft	Remove one set
400-600 km	April 2024 control	120 h pass / 144 h fail	120 h pass / 144 h fail	0/95 deletions change a reported criterion
400-600 km	May 2024	48 h pass / 72 h fail	48 h pass / 72 h fail	0/91 deletions change a reported criterion
400-600 km	October 2024 held out	24 h pass / 36 h fail	24 h pass / 36 h fail	0/100 deletions change a reported criterion
400-600 km	August 2024 held out	48 h pass / 72 h fail	36 h pass / 48 h fail; 48 h pass / 72 h fail	0/93 deletions change a reported criterion
600-750 km	April 2024 control	120 h pass / 144 h fail	120 h pass / 144 h fail; passes through the longest tested lead (168 h)	3/45 deletions change a reported criterion
600-750 km	May 2024	passes through the longest tested lead (168 h)	120 h last pass; unavailable at 144 h; passes through the longest tested lead (168 h)	0/41 deletions change a reported criterion
600-750 km	October 2024 held out	96 h pass / 120 h fail	96 h pass / 120 h fail; passes through the longest tested lead (168 h)	1/42 deletions change a reported criterion
600-750 km	August 2024 held out	120 h last pass; unavailable at 144 h	120 h last pass; unavailable at 144 h	0/39 deletions change a reported criterion
750-850 km	April 2024 control	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/57 deletions change a reported criterion
750-850 km	May 2024	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/65 deletions change a reported criterion
750-850 km	October 2024 held out	120 h pass / 144 h fail	48 h pass / 72 h fail; passes through the longest tested lead (168 h)	2/51 deletions change a reported criterion
750-850 km	August 2024 held out	passes through the longest tested lead (168 h)	144 h last pass; unavailable at 168 h; passes through the longest tested lead (168 h)	0/60 deletions change a reported criterion
850-1000 km	April 2024 control	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/78 deletions change a reported criterion
850-1000 km	May 2024	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/77 deletions change a reported criterion
850-1000 km	October 2024 held out	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/60 deletions change a reported criterion

Band	Window	Primary observed endpoints	Remove one spacecraft	Remove one set
850-1000 km	August 2024 held out	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/65 deletions change a reported criterion
1000-1400 km	April 2024 control	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/37 deletions change a reported criterion
1000-1400 km	May 2024	passes through the longest tested lead (168 h)	144 h pass / 168 h fail; passes through the longest tested lead (168 h)	19/30 deletions change a reported criterion
1000-1400 km	October 2024 held out	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/32 deletions change a reported criterion
1000-1400 km	August 2024 held out	passes through the longest tested lead (168 h)	passes through the longest tested lead (168 h)	0/35 deletions change a reported criterion

The low-altitude brackets survive deletion of any whole element set. Removing a spacecraft preserves the April 2024 control, May 2024 and October 2024 held out brackets; in August 2024 held out it can move the bracket earlier. The intermediate bands are more sensitive to who remains in the sample. In May 2024, the 600-750 km band passes through the longest tested lead with only 12 CryoSat-2 sets remaining. In August 2024 held out, its next lead has no usable observations. Above 850 km, the pooled bands pass through the longest tested lead, but the highest band in May 2024 can fail there after a spacecraft deletion.

Successive fits share tracking data; leads reuse trajectories; missions share orbit producers and storm conditions. The deletion checks describe fragility under those dependencies. They are not confidence intervals, and the selected disturbed windows cannot supply a sampling distribution for future storms.

September calendar replay

The pre-registered endpoint was signed in-track error 96 hours after the first post-manoeuvre element-set epoch. The locally frozen protocol yielded 13 complete recorded-burn predictor/endpoints, with mean absolute prediction error 4.284 km against 17.254 km for the zero-prediction baseline. The energy predictor beat that baseline in 11 of 13 events; SWOT and one Sentinel-3A event did not.

The failures matter. GRACE-FO D has a calibrated burn fraction of -2.397 and a 28.35 km prediction error. It still improves on its 55.20 km zero-baseline error, but that large residual limits the diagnostic. SWOT's event on 2024-09-13 has the wrong sign: the prediction is +2.232 km and the observed residual is -0.525 km. Its 2.757 km prediction error exceeds the 0.525 km baseline. The other baseline failure is Sentinel-3A on 2024-09-18, with a 1.037 km prediction error against 0.708 km for the zero baseline.

Calibration uses only earlier clean sets. Every published in-month burn remains in the inventory, including events with an intervening burn. The predeclared subset without another recorded burn has 12 complete endpoints. The mean prediction error stays below the zero baseline in all 7 spacecraft deletions.

The event inventory is not an independent sample. Its 13 rows share 12 distinct endpoints. Closely spaced Sentinel-3B burns overlap the fixed energy plateaus; the no-later-burn subset does not isolate the earlier plateau. An intermediate calibrated fraction therefore does not establish mixed catalogue fitting for an isolated manoeuvre.

The protocol and separate author attestation preceded experiment access locally; this is not externally timestamped preregistration. Earlier public-element and weather history and overlapping orbit products already existed. This replay is a retrospective physical diagnostic, not a prediction shown to have been available at element-set publication. September will not be reused as a hold-out for any new recipe. The [status page](#) and [complete event tables](#) retain chronology, exclusions, unresolved classifications and source-retention limits.

Component covariance and reference checks

Consistency covariance measures disagreement between catalogue solutions. It can miss shared error or include independent fit noise, so it bounds absolute accuracy in neither direction. We transport the predicted-state RIC covariance into each residual's truth RIC basis before extracting component sigmas. Component intervals are distinct from joint spatial ellipsoid coverage; this paper reports the former.

Population	Window	Lead h	Component	Inside 2 σ
400-600 km	April 2024 control	24	radial	95/95 (100.0%)
400-600 km	April 2024 control	24	in_track	93/95 (97.9%)
400-600 km	April 2024 control	24	cross	5/95 (5.3%)
Jason-3	April 2024 control	24	radial	14/19 (73.7%)
Jason-3	April 2024 control	24	in_track	19/19 (100.0%)
Jason-3	April 2024 control	24	cross	15/19 (78.9%)
Sentinel-6A	April 2024 control	24	radial	15/18 (83.3%)
Sentinel-6A	April 2024 control	24	in_track	0/18 (0.0%)
Sentinel-6A	April 2024 control	24	cross	16/18 (88.9%)

The table illustrates why coverage must be stated by component and mission. In-track intervals fail during storms, while even the control can fail in another component. At the same nominal high altitude, Jason-3 passes the 2 σ in-track check for 19/19 sets while Sentinel-6A fails completely, 0/18; the latter's fitted median sigma is 0.038 km, about a tenth of its 0.288 km median residual, with no cause established by this comparison.

The storm term is outside this paper's scope; its coefficient support, residuals and shift diagnostics remain in [the benchmark tables](#) and [storm-term analysis](#).

Topocentric crossing agreement

An orbital-component angle is not a sky-track displacement, and dividing in-track error by orbital speed is not a beam-crossing time error. We instead compare complete topocentric trajectories with a rotating observer and fixed celestial pointings against the measured MeerKAT beams. The cases are constructed pointings on previously inspected windows. They do not represent an observing schedule or detected interference.

Of 10,310 source-filtered rows, 112 have qualifying nominal visibility and complete curves. These yield 18,144 beam/pointing/observation cases from 94 element sets on 12 spacecraft. Nominal eligibility excludes reference-visible events whose predictions fail the gate. Repeated beams, offsets and intervals do not create independent trajectories.

For the full observation interval, the other declared offsets agree in 4,608/4,704 cases, with 45/2,016 false predicted crossings and 51/2,022 missed reference crossings. At the deliberately exact half-power boundary, agreement falls to 700/1,344. This construction-based split is a sensitivity check, separate from numerical near-boundary diagnostics.

Element-set age h	Pointings	Agreement / all cases	False / predicted crossings	Missed / reference crossings	Closest times n	Median absolute Δt s	Linear p95 absolute Δt s
6	Exact boundary offsets	30/60 (50.0%)	30/60	0/30	60	0.0652545	0.161698
6	Other declared offsets	210/210 (100.0%)	0/90	0/90	210	0.0652724	0.161719
12	Exact boundary offsets	228/420 (54.3%)	191/419	1/229	420	0.0596551	0.221621

Element-set age h	Pointings	Agreement / all cases	False / predicted crossings	Missed / reference crossings	Closest times n	Median absolute Δt s	Linear p95 absolute Δt s
12	Other declared offsets	1470/1470 (100.0%)	0/630	0/630	1470	0.0596575	0.22162
24	Exact boundary offsets	106/204 (52.0%)	98/204	0/106	204	0.058273	1.60689
24	Other declared offsets	691/714 (96.8%)	11/306	12/307	714	0.0583181	1.60688
36	Exact boundary offsets	6/12 (50.0%)	6/12	0/6	12	0.0541513	0.0544004
36	Other declared offsets	42/42 (100.0%)	0/18	0/18	42	0.0540956	0.0544217
48	Exact boundary offsets	164/324 (50.6%)	160/324	0/164	324	0.0225135	0.800852
48	Other declared offsets	1109/1134 (97.8%)	11/486	14/489	1134	0.0225488	0.800856
72	Exact boundary offsets	29/60 (48.3%)	30/59	1/30	60	0.0809106	4.8007
72	Other declared offsets	198/210 (94.3%)	6/90	6/90	210	0.0809555	4.80074
96	Exact boundary offsets	0/0 (unavailable)	0/0	0/0	0	unavailable	unavailable
96	Other declared offsets	0/0 (unavailable)	0/0	0/0	0	unavailable	unavailable
120	Exact boundary offsets	40/72 (55.6%)	32/72	0/40	72	0.946547	3.68884
120	Other declared offsets	229/252 (90.9%)	11/108	12/109	252	0.947119	3.68884
144	Exact boundary offsets	43/84 (51.2%)	41/84	0/43	84	0.181079	12.5503
144	Other declared offsets	281/294 (95.6%)	6/126	7/127	294	0.181252	12.5504
168	Exact boundary offsets	54/108 (50.0%)	54/108	0/54	108	0.0770706	0.623042
168	Other declared offsets	378/378 (100.0%)	0/162	0/162	378	0.0771282	0.622893

Each populated age bin has a different spacecraft composition, so these agreements do not establish an increase in agreement with element-set age.

Age bins are measured from element-set epoch, not publication. Agreement includes both crossing and non-crossing cases. Closest-time errors use all finite uncensored paired closest times, including false and missed crossings; entry and exit errors in the supplement require suitable crossings on both tracks. The table therefore keeps classification denominators beside timing errors. Receiver response, received power, coherence and real monitoring outcomes remain unmeasured. The measured beam data are CC BY-NC 4.0 and are used for research only.

Rejected learned propagators

The 2 local learned-propagator checkpoints lower the pooled paired median in 0/40 model/window/lead cells and lower the interpolated tail in 3/40. Both are rejected by the stored adoption rule. Some individual spacecraft improve while tails worsen. These are negative findings for the stated checkpoints, population and recipe, not a rejection of learned propagation as a field.

This is a local adaptation of the published dSGP4 method. Checkpoints were selected by the fixed training objective and checked again after reload. The evaluation windows were already inspected before correction; their historical names do not make this rerun confirmatory. Training traces, paired denominators, recipes and checkpoint identities remain in [the supplement](#).

Corrections and reproducibility

The covariance-basis correction dated 2026-09-08 changes coverage counts in 39/2400 original benchmark cells and 10/570 September secondary cells. All affected coverage counts are radial. Residuals, exclusions, in-track coverage counts and horizons are unchanged. [Every cell's change](#) is retained alongside the earlier manoeuvre-record and statistical corrections. The frozen September outputs remain intact; the supplement identifies the later covariance amendment.

The earlier post-manoeuve classifications describe calibrated energy compatibility, not the catalogue's unobserved fitting procedure. Their thresholds were author-reported as chosen before execution, but the first commit containing them also contained results. The later dated classifier freeze cannot establish earlier preregistration. State epoch spacing does not establish publication cadence.

The evidence object binds input files, method definitions, source hashes and completed-result metadata. Prose is authored in a Markdown template; only figures and tables are substituted from evidence. Tests reject unbound figures or stale rendered text. [Rebuild instructions](#) specify the retained inputs and checks. The [complete tables](#) hold hashes, checkpoints, timestamps and detailed references; the [archived earlier paper](#) preserves the superseded record.

Independent checks exposed clock, frame, fitting-window and station-reference-point errors that internally consistent calculations had missed. Those checks support the stated comparisons. They do not establish independent orbit determination, calibrated operational collision probabilities, warning completeness or a general radio operating horizon. Short-lead timing applications also require the open Cartesian dynamical validation recorded in the [roadmap](#).

Assurance appendix

Boundary	Fault	Independent check
Declared time system to comparison epoch	GPS orbit epochs were read as UTC.	Read the orbit header independently and compare lead-dependent residuals after conversion.
Provider states to propagation frame	SpaceX inertial states were interpreted in the catalogue frame; the header identified only the covariance frame.	Compare transformed states with CelesTrak's fit to the same source file, and check the rotation with a second library.
Sampled position to instantaneous velocity	A finite-difference chord biased the energy-derived semi-major axis.	Reproduce the bias on an orbit with known velocity and calibrate against earlier clean reference arcs.
Station ground marker to telescope reference point	Separately published laser-station eccentricities were omitted.	Compare residuals by station and elevation, using ILRS offsets independently of the orbit reader.
Configured fit window to actual inputs	A live covariance fit included stored sets outside its labelled window.	Count and bound the input epochs themselves, then compare the refitted result with the earlier fit.
Process exit status to completed validation	The density-model library could terminate the interpreter before the test runner finished while returning success.	Require a completed test report and compare its count with fresh collection and the retained test inventory.

The [convention notes](#), [frame comparison](#), [archived fault record](#) and [test-count guard](#) retain the supporting checks. The covariance-basis amendment is recorded separately in the corrections section above.

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Data and code availability

The [driftwatch repository](#) provides the code, evidence object, recorded tables and reproduction instructions. Release [paper-2026-09-v2](#) freezes the paper and supporting repository state; the Zenodo version DOI is [10.5281/zenodo.22661809](#). Code is distributed under the [MIT licence](#). Third-party inputs retain their source terms, and the code licence does not relicence them. Raw SpaceX ephemerides and ESA Kelvins inputs are not redistributed. Product identifiers, retained-file hashes, access limits and reconstruction commands are recorded in the [data-source record](#), [complete tables](#) and [rebuild instructions](#). Citation metadata is in [CITATION.cff](#).

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